Lumbre HB-320 Olive-Stone Boiler and Malaxer Hot-Water Loop: Commissioning Report

Almazara Torrebermeja S.C.A., Jaén · Maintenance and Energy Office · CR-CAL-2026-03 rev. 1 · Issued 21 October 2026 · Supersedes the draft rev. 0 of 17 October 2026

1. SCOPE AND INSTALLATION

This report records the cold and hot commissioning of the new olive-stone biomass boiler at the Almazara Torrebermeja S.C.A. mill, Carretera JA-3200 km 4, 23009 Jaén. The boiler is a **Lumbre HB-320** (nominal output 320 kW, serial number `HB320-26-0418`) supplied and installed by Calderas Guadalbullón S.L. of Mengíbar under order `PED-2026-117`. It replaces the 250 kW diesel boiler used up to and including the 2025/26 campaign, which burned on average 31,400 litres of heating oil per season.

The boiler heats a closed water loop that serves the six malaxers of Lines 1 and 2 (three per line, in the ground-floor process hall), the washer make-up water heater and the decanter dilution water tank. It burns the cooperative's own olive stone, separated from the pomace by the pitting unit in Building C and stored in the 18 t steel silo `SIL-04` (usable capacity 16.5 t). The loop design temperatures are 80 °C flow and 60 °C return.

- **Cold commissioning** (pressure test, interlocks, dry runs of the stone auger): 6 to 8 October 2026.
- **Hot commissioning** (first firing, efficiency, emissions and malaxer tests): 13 to 16 October 2026.
- **Hydrostatic test:** 6.0 bar for 2 h on 6 October, no drop; safety valve `PSV-12` set at 4.0 bar.
- **Witnesses:** Inmaculada Cobo Martos (commissioning engineer, Calderas Guadalbullón S.L.), Rafael Expósito Liébana (technical manager, Almazara Torrebermeja) and Francisco Jesús Moral Higueras (head miller, maestro de almazara).

2. MALAXER HEAT DEMAND

The boiler must cover the heat needed to bring the olive paste from the hammer-mill outlet temperature up to the malaxation set point. The cooperative markets all Line 2 oil as cold extracted, so the paste must never exceed 27 °C; the standing set point for both lines is 26 °C. The heat demand of one line is estimated with equation (1).

$$Q_m = \frac{\dot{m}_p \cdot c_p \cdot \left(T_s - T_{in} \right)}{3600 \cdot \eta_d} \tag{1}$$

where Q_m is the heat the boiler must deliver to one line (kW), $\dot{m}_p$ the paste flow (kg/h), c_p the specific heat of the paste, taken as 3.05 kJ/(kg·K) for the cooperative's Picual olives, T_s the malaxation set point (°C), T_{in} the paste temperature at the hammer-mill outlet (°C), η_d the distribution efficiency of the loop, measured at 0.88, and 3600 converts kJ/h to kW.

Worked example (Line 2, test of 15 October). Paste flow $\dot{m}_p$ = 6,200 kg/h, T_{in} = 11 °C, T_s = 26 °C. The heat taken up by the paste is 6,200 × 3.05 × 15 / 3,600 = 78.8 kW, and the boiler must deliver 78.8 / 0.88 = **89.5 kW** to Line 2.

> **Note.** The malaxer valves `TV-21` to `TV-26` close automatically when the paste probe reads 26.5 °C. During the 15 October test the highest paste temperature recorded on Line 2 was 26.3 °C, inside the cold-extraction limit.

3. FUEL QUALITY AND HEATING VALUE

The net heating value of the stone as received falls with moisture and is taken from equation (2); the dry value was measured on three samples from `SIL-04`.

$$H_w = H_0 \cdot \left(1 - w \right) - 0.678 \cdot w \tag{2}$$

where H_w is the net heating value as received (kWh/kg), H_0 the net heating value of dry stone, 5.35 kWh/kg, w the moisture content as a mass fraction (wet basis), and 0.678 kWh/kg the heat needed to evaporate the water contained in the stone.

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The test batch had w = 0.11 (11 %), giving H_w = 5.35 × 0.89 − 0.678 × 0.11 = **4.687 kWh/kg**. For comparison, fresh stone straight from the pitting unit is usually at 16 % to 20 % moisture (H_w = 4.39 to 4.14 kWh/kg).

Fuel rules for the boiler, agreed with the supplier for the warranty:

- Stone at or below 15 % moisture may be fed directly from `SIL-04`.
- Stone between 15 % and 20 % moisture may be burned only after blending 1:1 by volume with stone at or below 10 % moisture in the mixing hopper `TOL-2`, and the boiler output must then be limited to 70 % (224 kW).
- Stone above 20 % moisture must not enter the silo; it goes to the drying yard behind Building C for at least 10 days.

4. EFFICIENCY TEST

Efficiency was measured by the direct method, equation (3), with flow meter `FT-31`, probes `TT-31` / `TT-32` and the stone feed weighed over 30 minutes.

$$\eta_b = \frac{\dot{V} \cdot 1.163 \cdot \left(T_f - T_r \right)}{\dot{m}_s \cdot H_w} \tag{3}$$

where η_b is the boiler efficiency (fraction), $\dot{V}$ the loop water flow (m³/h), 1.163 kWh/(m³·K) the volumetric heat capacity of water at loop temperature, T_f and T_r the flow and return temperatures (°C), $\dot{m}_s$ the stone feed rate (kg/h) and H_w the net heating value from equation (2) (kWh/kg).

Acceptance criteria in the order: at least **86 %** at nominal load and at least **82 %** at part load (about 30 % of nominal output). The tender of March 2026 had asked for 88 % at nominal load; this was reduced to 86 % in contract amendment 2 of 12 May 2026.

- **Nominal load, 14 October:** $\dot{V}$ = 14.2 m³/h, T_f = 78.4 °C, T_r = 60.1 °C, output 302.2 kW; $\dot{m}_s$ = 73.8 kg/h, fuel input 345.9 kW; η_b = **87.4 %**. Passed.
- **Part load, first run, 14 October:** $\dot{V}$ = 5.1 m³/h, T_f = 74.9 °C, T_r = 58.6 °C, output 96.7 kW; $\dot{m}_s$ = 25.6 kg/h; η_b = 80.6 %. Failed, excess air too high.
- **Part load, repeat run, 16 October** (after the secondary air damper was re-set from 45 % to 30 % open): same flow and temperatures, $\dot{m}_s$ = 24.8 kg/h; η_b = **83.2 %**. Passed; this repeat run is the result of record.

5. EMISSIONS AND SAFETY CHECKS

Flue gas was measured at the chimney sampling port by Inmaculada Cobo Martos. At nominal load the results, corrected to 6 % O_2, were: CO 182 mg/Nm³ (limit 500), particulate 96 mg/Nm³ (limit 150) and NO_x 241 mg/Nm³ (limit 500). All values met the permit limits.

Safety functions passed: auger burn-back sprinkler (85 °C), over-temperature cut-out (95 °C), flame-loss stop, low-water cut-out and the emergency stop.

6. OPEN POINTS AND ACCEPTANCE

Provisional acceptance was granted on 16 October 2026 by Rafael Expósito Liébana. Category A points carry a restriction until closed; category B points are closed by the contractor within 30 days.

- **P-01 (B):** 6 m of loop pipework above the Line 1 malaxers not yet insulated.
- **P-02 (B):** silo level sensor `LT-04` reads 0.4 t high; recalibrate.
- **P-03 (A):** the GSM alarm dialler that calls the head miller's phone on a boiler fault is not yet connected. Until P-03 is closed (planned 6 November 2026), the boiler may run only while a trained operator is present in the mill; unattended night operation is not allowed, even with the mill closed.

Silo autonomy at nominal load and a 22-hour milling day is 16,500 / (73.8 × 22) = 10.2 days. The head miller orders a top-up from the pitting unit when the silo log shows less than 4 days of stone.

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